

Experiment No. 2

DIFFRACTION GRATING

Aim: To determine the wavelength of the spectral lines of the mercury spectrum using a diffraction grating.

Apparatus: Diffraction grating, spectrometer, mercury vapor lamp, spirit level, magnifying lens.

Formula:

Condition of maxima in diffraction grating experiment is $d \sin \theta = n\lambda$

$$\text{Hence, } \lambda = \frac{d \sin \theta}{n} \text{ m}$$

d = grating constant (m)

θ = angle of diffraction corresponding to wavelength λ and n is the order of spectrum

Grating Constant: The distance between two adjacent slits is known as grating constant. Its value can be obtained by dividing the length of grating by the total number of lines ruled on the grating or if number of lines per unit length is given then it will be reverse of that in that unit.

$$d = L / N;$$

L = length of the grating (m), usually on grating it is given in inches

N = number of lines ruled on the grating

Observation:

No. of lines per inch on grating is (N) = 15000,

$$1 \text{ inch} = 2.54 \times 10^{-2} \text{ m}$$

$$d = \frac{2.54 \times 10^{-2}}{15000} \text{ m} = 1.693 \times 10^{-6} \text{ m} = 1693 \text{ nm}$$

Principle:

A grating consists of a large number of parallel slits of very small width. When a parallel beam of white light is incident on a grating, Fraunhofer diffraction occurs. By measuring the angle of diffraction for different spectral lines, their wavelengths can be determined.

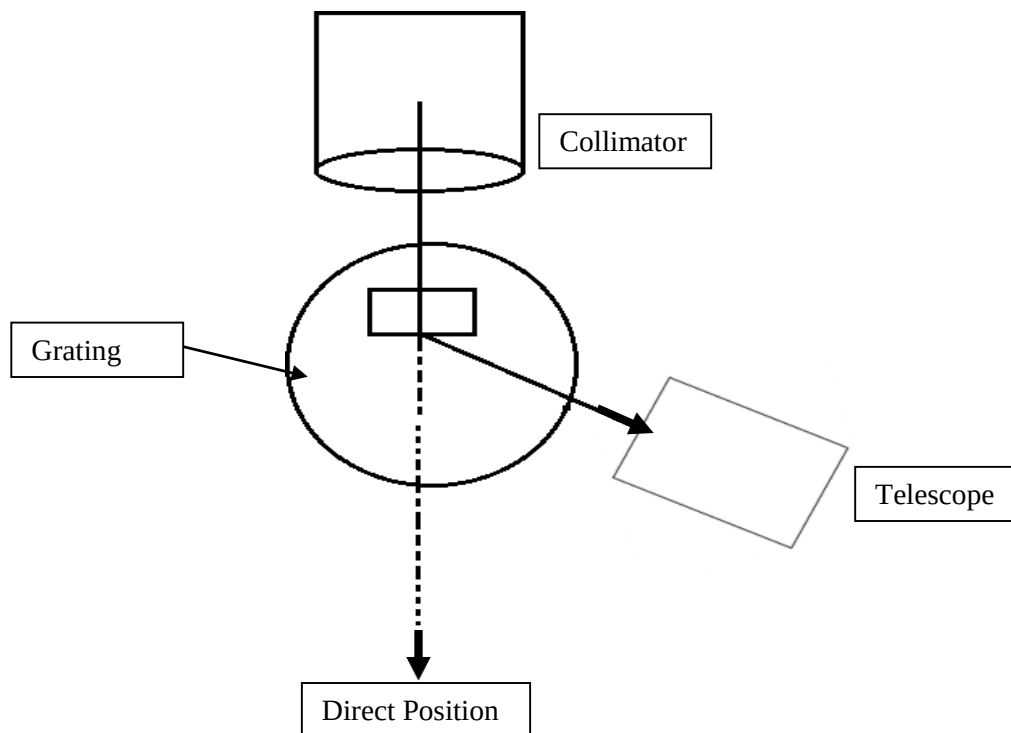
Procedure: Preliminary adjustments of the spectrometer

1. Adjust the eye-piece to see clear image of the crosswire.
2. The telescope is directed towards a distant object and the distance between the eyepiece and the objective lens is adjusted to get a clear image. The telescope now receives a parallel beam.
3. A spirit level is placed on the prism table and the screws at the base are adjusted till it is perfectly horizontal.
4. The Vernier table is fixed and the grating is mounted on the turn table and set for normal incidence.
5. The telescope is brought in line with the collimator and the direct reading R_0 is noted.
6. Now the telescope is turned to view the first order spectrum.
7. The vertical cross-wire is made to coincide with the colors and the respective diffraction angles 'R' are noted.
8. The difference between the direct reading and the diffracted reading for a particular colour gives the diffraction angle of the corresponding spectral line.
9. Using the calculated value of d , the wavelengths of the spectral lines are determined.

Setting the grating for normal incidence: -

To set the grating in the normal incidence position, the telescope is brought in line with the collimator to have image of the slit on the vertical cross wire. It is locked in that position. The lower screw of the spectrometer is unlocked and the table is rotated till the reading on one vernier is exactly 360 degrees. The reading on the second vernier is therefore 180 degrees. The screw is locked. Now the telescope arm is unlocked and rotated through 90 degrees in clockwise or anticlockwise direction. It is locked in that position. The grating stand is mounted on the prism table and the grating is placed in it. Care is taken that the grating stands at the centre of the prism table. Then the prism table is rotated slowly so that a reflected image is seen in the field of view of the telescope. Let the image coincide with the vertical cross wire. In this position, the angle of incidence of light on the grating surface is 45 degrees. The reading of the position of the prism table is noted down. The prism table is rotated from the above position through 45 degrees more so that the grating plane becomes normal to the direction of light. The prism table is locked in this position. This is the normal incidence position.

Diagram:



Spectrometer Readings:

$$\text{Least Count (L.C.)} = \frac{\text{Value of 1MSD (in degree)}}{\text{No. of divisions on V.S.}} \quad \text{L.C.} = \text{-----degree} = \text{----- minute}$$

$$\text{Direct Reading, } R_o = \text{MSR} + (\text{VSD} \times \text{L.C.}) = \text{----- degree; Order of spectrum } n = 1$$

Sr.No	Color	MSR Degree	VSD	TR degree	$\theta = R - R_o$ degree	$\lambda = \frac{d \sin \theta}{n}$ nm
1	Violet					
2	Green					
3	Yellow					
4	Red					

MSR = Main Scale Reading, VSD = Coinciding Vernier Scale Division

$$\text{TR} = \text{Total Reading} = \text{MSR} + (\text{VSD} \times \text{L.C.})$$

- Calculation:**
1. Grating constant $d = \text{----- (m)}$
 2. Calculate the wavelengths of different spectral lines and compare with the standard values given in above table.

$$\lambda = \frac{d \sin \theta}{n} \text{ nm}$$

Result:

Color	Wavelength λ (nm)
Violet	
Green	
Yellow	
Red	

Standard Wavelengths of Spectral Lines

Color	Wavelength λ (nm)	Color	Wavelength λ (nm)
Bright Violet	404.7 nm	Dull Green	491.6 nm
Violet	407.8 nm	Bright Green	546.1 nm
Blue	435.8 nm	Yellow	577.0 nm
Dull Green	491.6 nm	Yellow	579.1 nm

Precaution:

- 1.The grating should not be touched with hand or rubbed. It should always be held by means of fingers kept on the opposite edges of the grating.
- 2.Grating should be perfectly normal to the axis of the collimator.
- 3.The turn table must be leveled optically.
- 4.The slit should be as narrow as possible.
- 5.All the preliminary adjustments of the spectrometer must be made before starting the experiment.
6. While taking observations the turn table must remain clamped.